

Statistical Modeling of Narrowband Processes for Realistic Signal Representation

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August 8, 2026

- 1 Background
- 2 Filtering Random Processes and Spectral Factorization
- 3 Special Types of Random Processes
 - Autoregressive Moving Average (ARMA) Process
 - Autoregressive (AR) Process
 - Moving Average (MA) Processes
- 4 Narrowband Signal Modeling using Autoregressive Band-pass Processes
- 5 Algorithmic Validation
- 6 References

Content adapted from:

M. H. Hayes, *Statistical Digital Signal Processing and Modeling*, Wiley, 1996. [1]

Autocorrelation and Autocovariance Matrices

Signal Vector:

$$\mathbf{x} = [x(0), x(1), \dots, x(p)]^T$$

Autocorrelation Matrix:

$$\mathbf{R}_x = \mathbb{E}[\mathbf{x}\mathbf{x}^H] = \begin{bmatrix} r_x(0) & r_x^*(1) & \cdots & r_x^*(p) \\ r_x(1) & r_x(0) & \cdots & r_x^*(p-1) \\ \vdots & \vdots & \ddots & \vdots \\ r_x(p) & r_x(p-1) & \cdots & r_x(0) \end{bmatrix}$$

- For WSS processes: $r_x(k) = r_x^*(-k)$
- Hermitian Toeplitz structure

Autocovariance Matrix:

$$\mathbf{C}_x = \mathbb{E}[(\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^H]$$

Relation:

$$\mathbf{C}_x = \mathbf{R}_x - \mathbf{m}_x \mathbf{m}_x^H$$

$$\mathbf{m}_x = [m_x, m_x, \dots, m_x]^T$$

Eigenvalue Extremal Property

Statement:

- For a zero-mean WSS random process, the eigenvalues of the $n \times n$ autocorrelation matrix $\mathbf{R}_x$ are bounded by the extrema of the power spectral density (PSD).

$$\min_{\omega} P_x(e^{j\omega}) \leq \lambda_i \leq \max_{\omega} P_x(e^{j\omega})$$

- λ_i : eigenvalues of $\mathbf{R}_x$
- $P_x(e^{j\omega})$: power spectral density (PSD)

For zero-mean processes: $\mathbf{C}_x = \mathbf{R}_x$

Filtering Random Processes

System Model:

$$x(n) \xrightarrow{h(n)} y(n)$$

- $h(n)$: Linear Shift-Invariant (LSI) system

Autocorrelation Relationship:

$$r_y(k) = r_x(k) * h(k) * h^*(-k)$$

Power Spectral Density (PSD):

$$P_y(e^{j\omega}) = P_x(e^{j\omega}) |H(e^{j\omega})|^2$$

Z-domain Representation:

$$P_y(z) = P_x(z) H(z) H^*\left(\frac{1}{z^*}\right)$$

Spectral Factorization of PSD

Factorization:

$$P_x(z) = \sigma_0^2 Q(z) Q^* \left(\frac{1}{z^*} \right)$$

Normalization Constant:

$$\sigma_0^2 = \exp \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} \ln P_x(e^{j\omega}) d\omega \right)$$

- $Q(z)$: minimum-phase (all zeros inside unit circle)
- $P_x(e^{j\omega}) \geq 0$, real-valued

Any valid PSD can be expressed as a product of a filter and its conjugate-reciprocal.

Regular Process:

- A process with spectral factorization is called *regular*

Model:

$$x(n) = h(n) * v(n)$$

- $v(n)$: white noise with variance σ_0^2
- $h(n)$: causal, stable filter

Whitening Property:

$$v(n) = \frac{1}{H(z)} x(n)$$

- Inverse filter produces white noise

Any regular process can be realized as the output of a causal stable filter that is driven by white noise having variance of σ_o^2 .

Rational PSD and Wold Decomposition

Rational Spectrum:

$$P_x(z) = \sigma_0^2 \frac{B(z)}{A(z)} \cdot \frac{B^*\left(\frac{1}{z^*}\right)}{A^*\left(\frac{1}{z^*}\right)}$$

- $A(z), B(z)$: all roots inside unit circle
- Zeros/poles occur in conjugate-reciprocal pairs

Wold Decomposition:

A general random process can be written as a sum of two processes:

$$x(n) = x_r(n) + x_p(n)$$

- $x_r(n)$: regular (stochastic) component
- $x_p(n)$: predictable (deterministic) component

$$P_x(e^{j\omega}) = P_r(e^{j\omega}) + \sum_k \alpha_k \delta(\omega - \omega_k)$$

Special Types of Random Processes

- Many practical processes are obtained by filtering white noise
- Consider LSI systems with **rational transfer functions**

$$x(n) = h(n) * v(n)$$

- $v(n)$: white noise input
- $x(n)$: WSS output process

Classes of Processes:

- Autoregressive (AR)
- Moving Average (MA)
- Autoregressive Moving Average (ARMA)

Structured stochastic processes arise from filtering white noise through rational systems.

Autoregressive Moving Average (ARMA) Process

System Function:

$$H(z) = \frac{B_q(z)}{A_p(z)} = \frac{\sum_{k=0}^q b_k z^{-k}}{1 + \sum_{k=1}^p a_k z^{-k}}$$

Output PSD:

$$P_x(e^{j\omega}) = \sigma_v^2 \frac{|B_q(e^{j\omega})|^2}{|A_p(e^{j\omega})|^2}$$

- p : number of poles (AR part)
- q : number of zeros (MA part)

Processes with this particular characteristic of the power spectrum are called ARMA(p, q) processes.

Yule–Walker Equations

Relationship between filter coefficients and autocorrelation sequence

ARMA Difference Equation:

$$x(n) + \sum_{i=1}^p a_i x(n-i) = \sum_{l=0}^q b_l v(n-l)$$

Autocorrelation Relation:

$$r_x(k) + \sum_{i=1}^p a_i r_x(k-i) = \sum_{l=0}^q b_l r_{vx}(k-l)$$

For white noise input $v(n)$:

$$r_{vx}(k-l) = \sigma_v^2 h^*(l-k)$$

Yule–Walker Form:

$$r_x(k) + \sum_{i=1}^p a_i r_x(k-i) = \begin{cases} \sigma_v^2 c_q(k), & 0 \leq k \leq q \\ 0, & k > q \end{cases}$$

Where:

$$c_q(k) = \sum_{l=0}^q b_l h^*(l-k)$$

Yule–Walker in Matrix Form

$$\begin{bmatrix} r_x(0) & r_x(-1) & \cdots & r_x(-p) \\ r_x(1) & r_x(0) & \cdots & r_x(-p+1) \\ \vdots & \vdots & \ddots & \vdots \\ r_x(p) & r_x(p-1) & \cdots & r_x(0) \end{bmatrix} \begin{bmatrix} 1 \\ a_1 \\ \vdots \\ a_p \end{bmatrix} = \sigma_v^2 \begin{bmatrix} c_q(0) \\ c_q(1) \\ \vdots \\ c_q(p) \end{bmatrix}$$

Autoregressive (AR) Process

Definition (AR(p)):

- Special case of ARMA with $q = 0$ (no MA part)

System Function:

$$H(z) = \frac{b(0)}{1 + \sum_{k=1}^p a_k z^{-k}}$$

Power Spectral Density:

$$P_x(e^{j\omega}) = \sigma_v^2 \frac{|b(0)|^2}{|A_p(e^{j\omega})|^2}$$

- All-pole model $\rightarrow$ only poles shape the spectrum

Yule–Walker Equations (AR case):

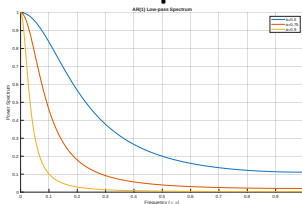
$$r_x(k) + \sum_{i=1}^p a_i r_x(k-i) = \sigma_v^2 |b(0)|^2 \delta(k)$$

AR(1) Processes

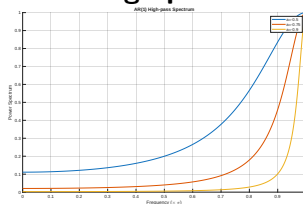
Model:

$$H(z) = \frac{1}{1 - az^{-1}}, \quad P_x(e^{j\omega}) = \frac{\sigma_v^2}{|1 - ae^{-j\omega}|^2}$$

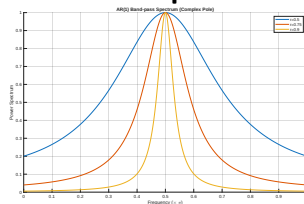
Low-pass



High-pass



Band-pass



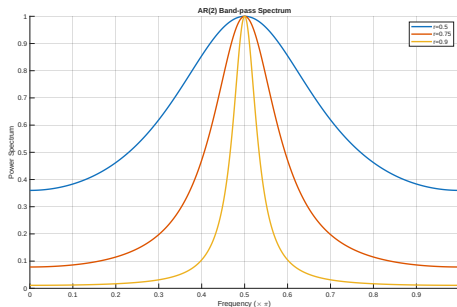
- Low-pass: $a = 0.5$, $a = 0.75$, $a = 0.9$, $\omega_0 = 0$
- High-pass: $a = -0.5$, $a = -0.75$, $a = -0.9$, $\omega_0 = \pi$
- Band-pass: $r = 0.5$, $r = 0.75$, $r = 0.9$, $\omega_0 = \pi/2$

AR(2) Band-pass Process

Model:

$$H(z) = \frac{1}{(1 - az^{-1})(1 - a^*z^{-1})} \quad , \quad a = re^{j\omega_0}$$

$$P_x(e^{j\omega}) = \frac{\sigma_v^2}{|1 - 2r \cos(\omega_0)e^{-j\omega} + r^2e^{-j2\omega}|^2}$$



• $r = 0.5$, $r = 0.75$, $r = 0.9$ ($\omega_0 = \pi/2$)

Moving Average (MA) Processes

Model:

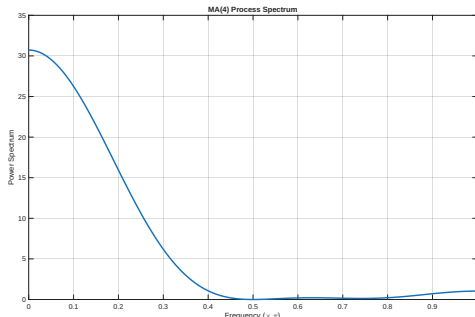
$$x(n) = \sum_{k=0}^q b_k v(n-k)$$

$$H(z) = \sum_{k=0}^q b_k z^{-k} \quad (\text{FIR})$$

Power Spectrum:

$$P_x(e^{j\omega}) = \sigma_v^2 |B_q(e^{j\omega})|^2$$

MA processes exhibit smoothly varying spectra with nulls determined by the zeros of $H(z)$. Zeros near the unit circle produce sharp spectral dips, while the spectrum varies more gradually compared to AR processes.



Signal Modeling for Narrowband Processes

Goal:

- Model narrowband signals for algorithm evaluation

Two equivalent representations:

- Real sinusoid:

$$s(n) = A \cos(\omega_0 n)$$

- Spectrum has peaks at $\pm\omega_0$
- Requires conjugate pole pair
- modeled using **AR(2)**

- Complex exponential:

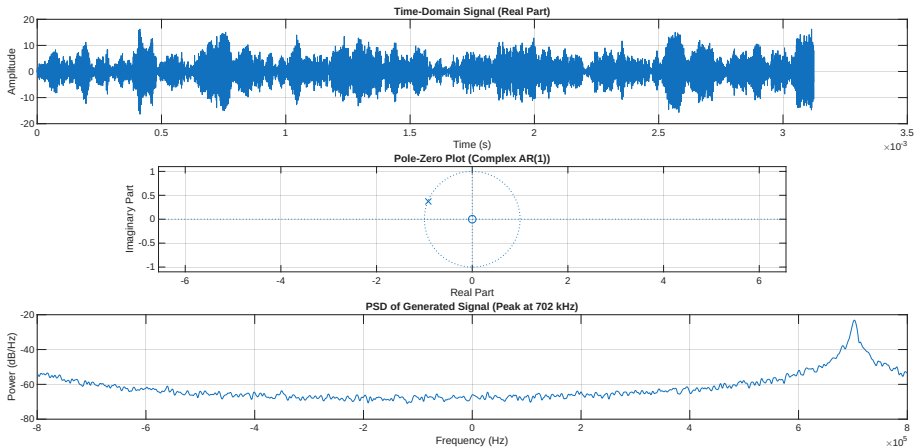
$$s(n) = Ae^{j\omega_0 n}$$

- Single spectral peak at ω_0
- Single pole representation
- modeled using **AR(1)**

Narrowband Signal Modeling (Complex AR(1))

Model:

- Center frequency: $f_0 = 702$ kHz, Sampling frequency: $f_s = 1.6$ MHz



A single spectral peak at $+f_0$ is observed, consistent with a complex narrowband process.

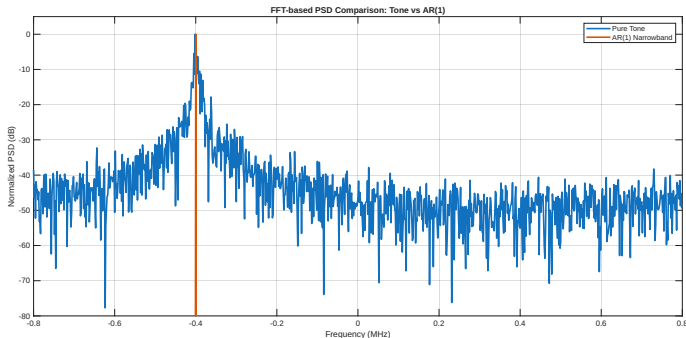
From Ideal to Realistic Signal Modeling

Initial Model (Ideal):

- Pure tone: $s(n) = Ae^{j\omega_0 n}$
- Single frequency component, infinitely narrow spectrum

Realistic Model:

- Narrowband random process (AR(1))
- Finite bandwidth around ω_0 , models practical signal variations



Pure tones are idealized; narrowband stochastic models better represent real-world signals.

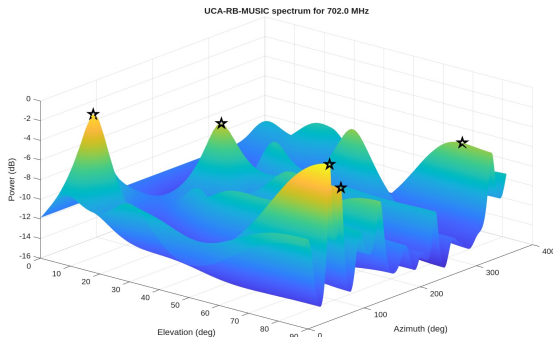
Algorithm Evaluation: LOS Estimation

Ground Truth

Src	$\phi(^{\circ})$	P (dB)
1(LOS)	43	-3
2	65	0
3	85	-12
4	100	-12

Detected Peaks

Pk	$\phi(^{\circ})$	P (dB)
1	44	-0.25
2	46	-1.63
3	64	-3.08
4	190	-4.35
5	333	-5.08



References I



M. H. Hayes, *Statistical Digital Signal Processing and Modeling*, 1st ed. USA: John Wiley & Sons, 1996.